



McLaren
USER MANUAL
MCLAREN 720S GT3



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DEAR iRACING USER,

Congratulations on your purchase of the McLaren 720S GT3! From all of us at iRacing, we appreciate your support and your commitment to our product. We aim to deliver the ultimate sim racing experience, and we hope that you'll find plenty of excitement with us behind the wheel of your new car!

The following guide explains how to get the most out of your new car, from how to adjust its settings off of the track to what you'll see inside of the cockpit while driving. We hope that you'll find it useful in getting up to speed.

Thanks again for your purchase, and we'll see you on the track!



CHASSIS

**DOUBLE-WISHBONE FRONT
AND REAR SUSPENSION**



LENGTH
4664 mm
183.6 in

WIDTH
2040 mm
80.3 in

WHEELBASE
2696 mm
106.1 in

DRY WEIGHT
1300 kg
2932 lbs

WET WEIGHT
WITH DRIVER
1494 kg
3293 lbs

POWER UNIT



TWIN-TURBOCHARGED, FLAT-PLANE V8

DISPLACEMENT
4.0 Liters
244 CID

RPM LIMIT
8000 RPM

TORQUE
492 lb-ft
667 Nm

POWER
520 bhp
387 kW



INTRODUCTION

The information found in this guide is intended to provide a deeper understanding of the chassis setup adjustments available in the garage, so that you may use the garage to tune the chassis setup to your preference.

Before diving into chassis adjustments, though, it is best to become familiar with the car and track. To that end, we have provided baseline setups for each track commonly raced by these cars.

To access the baseline setups, simply open the Garage, click iRacing Setups, and select the appropriate setup for your track of choice. If you are driving a track for which a dedicated baseline setup is not included, you may select a setup for a similar track to use as your baseline.

After you have selected an appropriate setup, get on track and focus on making smooth and consistent laps, identifying the proper racing line and experiencing tire wear and handling trends over a number of laps.

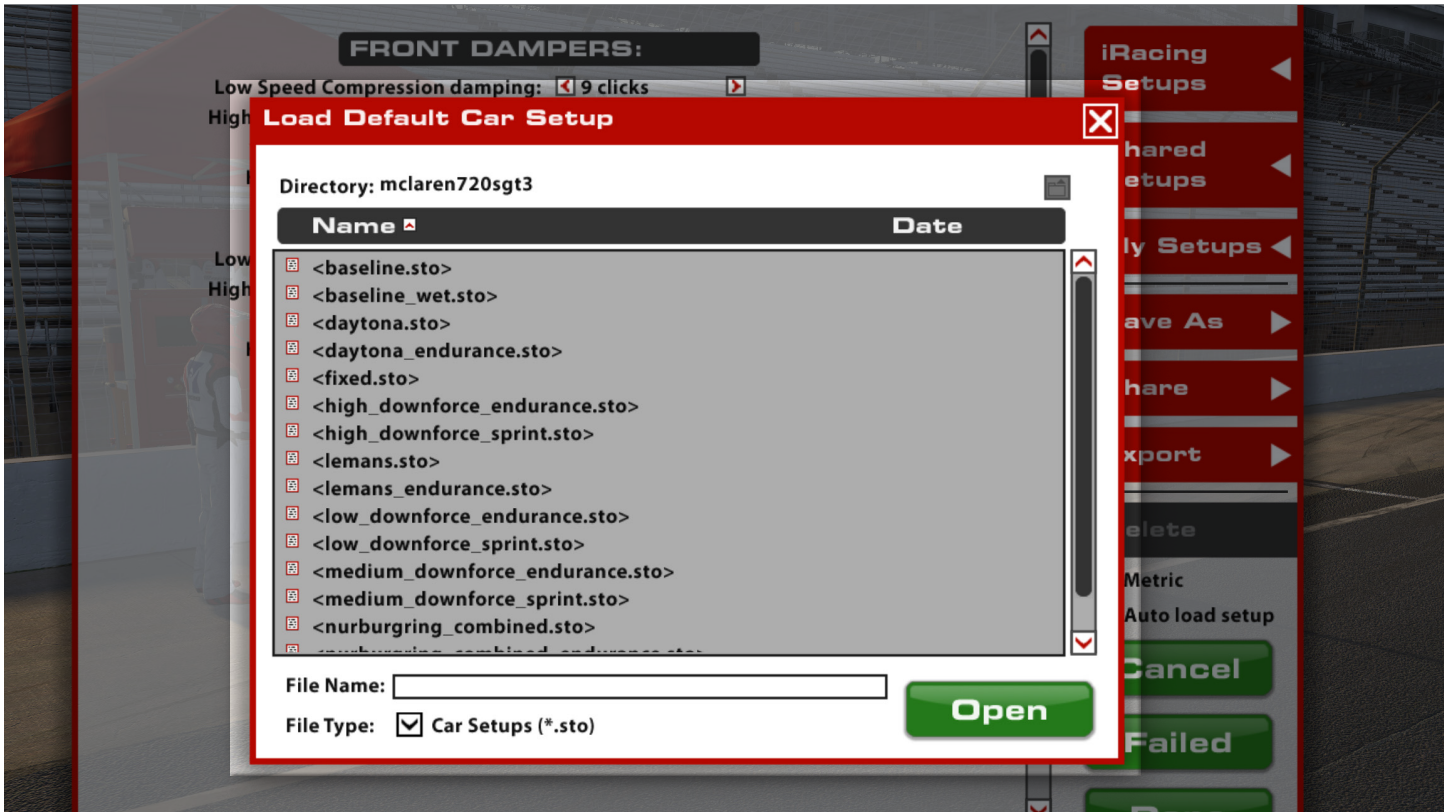
GETTING STARTED



Once you load into the car, getting started is as easy as selecting the “upshift” button to put it into gear, and hitting the accelerator pedal. This car uses a sequential transmission and does not require a clutch input to shift in either direction.

However the car’s downshift protection will not allow you to downshift if it feels you are traveling too fast for the gear selected and would incur engine damage. If that is the case, the gear change command will simply be ignored.

LOADING AN iRACING SETUP



Upon loading into a session, the car will automatically load the iRacing Baseline setup [baseline.sto]. If you would prefer one of iRacing's pre-built setups that suit various conditions, you may load it by clicking Garage > iRacing Setups > and then selecting the setup to suit your needs.

If you would like to customize the setup, simply make the changes in the garage that you would like to update and click apply.

If you would like to save your setup for future use click "Save As" on the right to name and save the changes. To access all of your personally saved setups, click "My Setups" on the right side of the garage.

If you would like to share a setup with another driver or everyone in a session, you can select "Share" on the right side of the garage to do so.

If a driver is trying to share a setup with you, you will find it under "Shared Setups" on the right side of the garage as well.

DASH CONFIGURATION

The McLaren 720S GT3 EVO features a three-page digital display mounted in the dash to provide information to the driver as clearly as possible. The default display page can be set in the garage and the page can be changed from within the car via the In-Car Adjustments black box.

RACE 1 DASH CONFIGURATION



Top Row	
Map	Current Throttle Shape Setting
TC1 & TC2	Current Traction Control system setting. These values are linked and will show the same value
ABS	Current Anti-Lock Brake System setting
FUNC	Inoperative
Center	
RPM	Engine RPM is shown at the top of the center column
Gear	The currently selected gear is shown in the center of the display
Speed	Vehicle speed, in kph or mph, is shown beneath the gear indicator
Left Cluster	
Predicted Laptime	The estimated lap time for the current lap, shown live, based on previous laps and current time delta.
Lap Delta	Current time difference between the current lap and the previous best lap time
Last Lap	The previously completed lap time
Laps	Number of laps completed since leaving the garage
Water Temp	Engine cooling water temperature (°C or °F). This value is color-coded: Blue is cold, white is optimum operating temperature, and red is overheated.
Battery Volts	Current electrical system voltage
Right Cluster	
Tire Pressure Text	Within the “Tyre Data” section, the live Tire Pressures are shown as the upper number for each corner of the car. These values are color-coded: Blue is low pressure indicating a new, cold tire, White is optimum pressure, Red is over-inflated. In the center of the Tire Data cluster are four blocks serving as visual representations of the Tire Pressures, Blue is a cold, low-pressure tire, Green is optimum pressure, Orange indicates the high pressure of a hot tire.
Tire Carcass Temperatures	Live tire carcass temperatures are shown beneath the pressures, color-coded in the same way as the pressures: Blue is too cold, White is optimum, and Red is over-heated.
Fuel Used	Amount of fuel used since leaving the pits
Fuel Level	Amount of fuel currently in the fuel tank
Fuel Last Lap	Amount of fuel used on the previous lap



RACE 2 DASH CONFIGURATION



The Race2 page is similar to the Race1 page, however the cluster of information on the right displays information about the braking system.

Right Cluster

Brake Temperature	The brake rotor temperatures are shown on the right side, in °C or °F, in color-coded values: Blue is cold, white is optimum, and red is overheated.
Tire Pressure Icons	As with Race1, the tire pressures are shown as color-coded blocks to convey when the tires are at optimum pressures or are under- or over-inflated.
Brake Balance F	The current brake bias setting, in % to the front, is shown below the Brake Temperatures and Tire Pressures.

QUALI DASH CONFIGURATION



The QUALI page strips away most of the information from the race page and gives the driver only the essentials needed for Qualifying sessions. The upper and center sections are the same as the Race pages, however the left and right clusters change.

Left Cluster

Time Delta Bar	The current time delta is shown as both a graphical bar and a value beneath the bar, comparing the current lap against the fastest lap of the session.
Last Laptime	The lap time for the previously completed lap

Right Cluster

Predicted Laptime	The estimated lap time for the current lap, shown live, based on previous laps and current time delta.
Tire Pressures	The live Tire Pressures are shown for each corner of the car below the Predicted Lap time. These values are color-coded: Blue is under-inflated, white is optimum pressure, red is over-inflated.

PIT SPEED LIMITER



Whenever the Pit Speed Limiter is activated the display will automatically switch to a dedicated screen to assist with pit road entry and maintaining pit road speed.

Dash Information	
Speed	The car's speed is shown at the top of the screen in the center
Tire Pressures	Tire pressures are shown below the speed in the center of the display
Gear	The currently-selected gear is shown at the bottom-center
Brake Balance F	This shows the current brake bias setting
Fuel Used	The amount of fuel used since the last time the car left the pits
Pitlane Timer	Time spent on pit road since the limiter was activated
Coolant Temp	Engine cooling water temperature (°C or °F). This value is color-coded: Blue is cold, white is optimum operating temperature, and red is overheated.
Screen Color	The Pit Limiter Screen will change color based on how fast the vehicle is traveling in relation to the pit road speed limit for the current track. If the speed is much faster than the pit road speed limit the screen will be black, changing to red when the speed is 10-20kph above the limit, yellow when the speed is 3-10kph above the limit, and green when the speed matches the pit road speed limit.

SHIFT & STATUS LIGHTS

Surrounding the digital display is a series of LED lights to quickly convey information about car performance and assist with shifting.



When the optimum RPM for an upshift is approaching, the entire set of LEDs will illuminate in blue. Drivers should initiate the upshift when these solid blue lights appear.



The goal is for the shift to occur when the blue/red LEDs appear (driver reaction time considered).



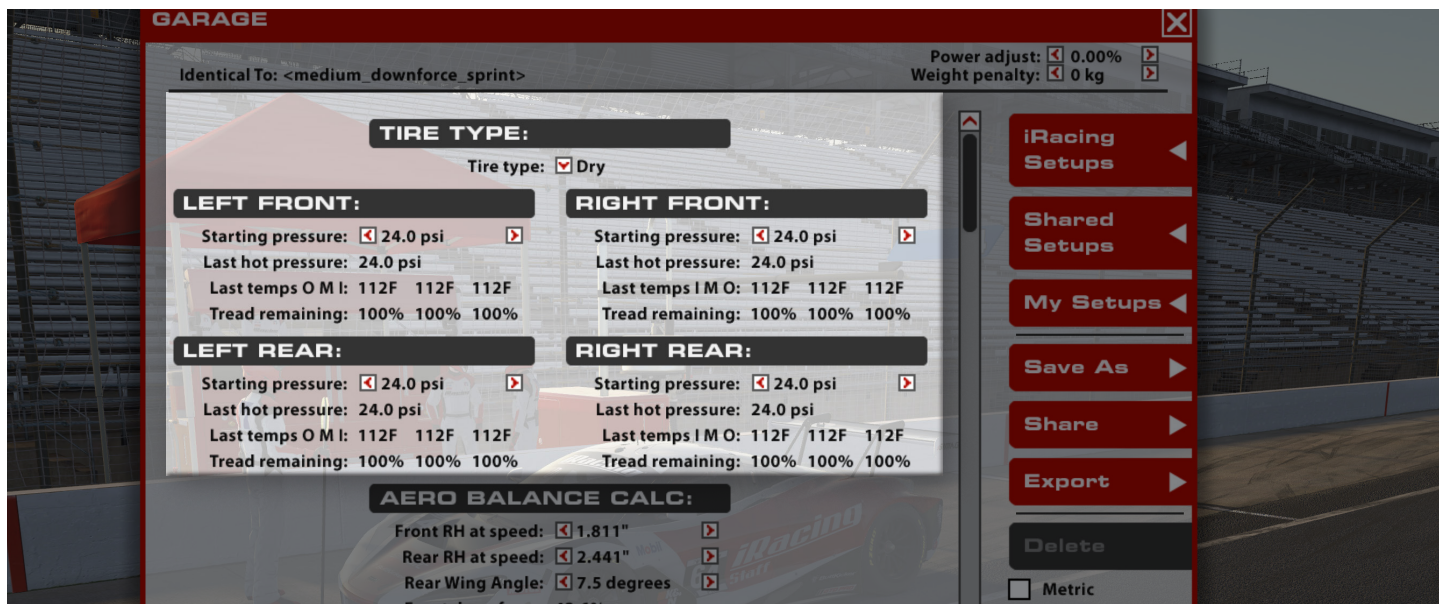
In the event of wheel lockup, LEDs on the side of the display will illuminate to represent which wheel is locking. Front wheels are indicated by yellow LEDs and rear wheels are indicated by green LEDs. Whenever the Traction Control system is on and attempting to reduce wheelspin, a blue LED will illuminate on the left side of the display.

ADVANCED SETUP OPTIONS

This section is aimed toward more advanced users who want to dive deeper into the different aspects of the vehicle's setup. Making adjustments to the following parameters is not required and can lead to significant changes in the way a vehicle handles. It is recommended that any adjustments are made in an incremental fashion and only singular variables are adjusted before testing changes.

TIRES & AERO

TIRE DATA



TIRE TYPE

Tires fitted to the McLaren 720S GT3 car can be changed based on weather conditions. The Dry option fits a slick tire intended for dry track conditions while the Wet option fits a treaded tire for wet track surfaces.

COLD PRESSURE / STARTING PRESSURE

The air pressure in the tires when the car is loaded into the world. Lower pressures will provide more grip but will produce more rolling drag and build temperature faster. Higher pressures will feel slightly more responsive and produce less rolling drag, but will result in less grip. Generally, higher pressures are preferred at tracks where speeds are higher while lower pressures work better at slower tracks where mechanical grip is important.

LAST HOT PRESSURE

When the car returns to the garage after an on-track stint, the tire pressure will be displayed as Hot Pressure. The difference between cold and hot pressure is a good way to see how tires are being loaded and worked while on track. Tires seeing more work will build more pressure, and paying attention to which tires are building more pressure and adjusting cold pressure to compensate can be crucial for optimizing tire performance.

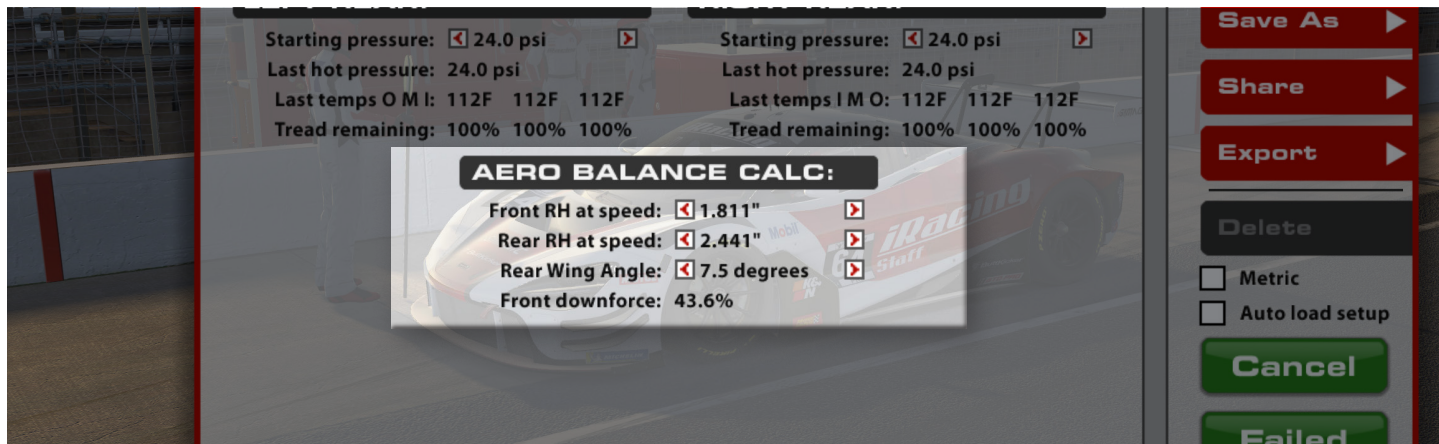
LAST TEMPS

The tire carcass temperatures (measured within the tread) are displayed after the car returns from the track. These temperatures are an effective way to determine how much work or load a given tire is experiencing while on track. Differences between the inner and outer temperatures can be used to tune individual wheel alignment and the center temperatures can be compared to the outer temperatures to help tune tire pressure.

TREAD REMAINING

The amount of tread on the tire, displayed as a percentage of a new tire, is shown below the tire temperatures. These values are good for determining how far a set of tires can go before needing to be replaced, but don't necessarily indicate an under- or over-worked tire in the same way temperatures will.

AERO BALANCE CALCULATOR



The Aero Calculator is a tool provided to aid in understanding the shift in aerodynamic balance associated with adjustment of the rear wing setting and front and rear ride heights. It is important to note that the values for front and rear ride height displayed here DO NOT result in any mechanical changes to the car itself, however, changes to the rear wing angle here WILL be applied to the car. This calculator is a reference tool ONLY.

FRONT RH AT SPEED

The Ride Height (RH) at Speed is used to give the Aero Calculator heights to reference for aerodynamic calculations. When using the aero calculator, determine the car's Front Ride height via telemetry at any point on track and input that value into the "Front RH at Speed" setting. It is advisable to use an average value of the LF and RF ride heights as this will provide a more accurate representation of the current aero platform rather than using a single corner height.

REAR RH AT SPEED

The Ride Height (RH) at Speed is used to give the Aero Calculator heights to reference for aerodynamic calculations. When using the aero calculator, determine the car's Rear Ride height via telemetry at any point on track and input that value into the "Front RH at Speed" setting. It is advisable to use an average value of the LR and RR ride heights as this will provide a more accurate representation of the current aero platform rather than using a single corner height.

REAR WING ANGLE

The Rear Wing Angle refers to the relative angle of attack of the rear wing, this is a powerful aerodynamic device which has a significant impact upon the total downforce (and drag) produced by the car as well as shifting the aerodynamic balance of the car rearwards with higher settings. Increasing the rear wing angle results in more total cornering grip capability in medium to high speed corners but will also result in a reduction of straight line speed. Rear wing angle should be adjusted in conjunction with front and rear ride heights, specifically the difference between front and rear ride heights known as "rake". To retain the same overall aerodynamic balance it is necessary to increase the rake of the car when increasing the rear wing angle.

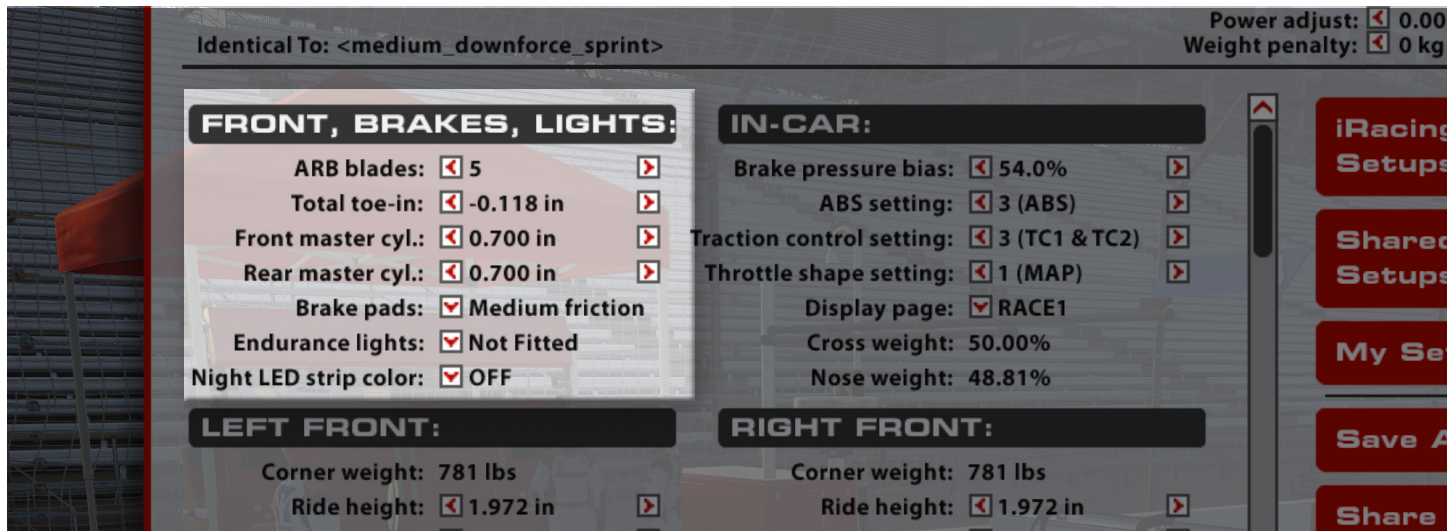
The Wing Angle value in the Aero Calculator section is tied directly to the Wing Angle in the Chassis page's Rear section. Changing one will automatically change the other. Information on how much to adjust ride heights for a given wing angle change can be found in the Setup Tips section of this guide.

FRONT DOWNFORCE

This value displays the proportion of downforce acting at the front axle for the given wing and ride height combination set within the calculator parameters. This value is an instantaneous representation of your aero balance at this exact set of parameters and it can be helpful to pick multiple points around a corner or section of track to understand how the aerodynamic balance is moving in differing situations such as braking, steady state cornering and accelerating at corner exit. A higher forwards percentage will result in more oversteer in mid to high speed corners.

CHASSIS

FRONT



ARB BLADES

The ARB Blades (or arms) can be adjusted to further tune the suspension roll stiffness beyond only the ARB size setting. This option changes the orientation of the ARB blades and are given numerical values for simplicity, with #1 being the softest option and the blades becoming stiffer as the value is decreased to the minimum setting of #8. Stiffer blade settings will increase front roll stiffness and induce understeer while softer blade settings will reduce front roll stiffness and reduce understeer.

TOTAL TOE-IN

Toe is the angle of the wheel, when viewed from above, relative to the centerline of the chassis. Toe-in is when the front of the wheels are closer to the centerline than the rear of the wheels, and Toe-out (Negative value in the garage) is the opposite. On the front end, adding toe-out will increase slip in the inside tire and decrease straight-line stability while adding toe-in will reduce the slip and increase straight-line stability.

FRONT MASTER CYLINDER

The Front Brake Master Cylinder size can be changed to alter the line pressure to the front brake calipers. A larger master cylinder will reduce the line pressure to the front brakes, which will shift the brake bias rearwards and increase the pedal effort required to lock the front wheels. A smaller master cylinder will increase brake line pressure to the front brakes, shifting brake bias forward and reducing required pedal effort to lock the front wheels.

REAR MASTER CYLINDER

The Rear Brake Master Cylinder size can be changed to alter the line pressure to the rear brake calipers. A larger master cylinder will reduce the line pressure to the rear brakes, which will shift the brake bias forwards and increase the pedal effort required to lock the rear wheels. A smaller master cylinder will increase brake line pressure to the rear brakes, shifting brake bias rearward and reducing required pedal effort to lock the rear wheels.

BRAKE PADS

The vehicle's braking performance can be altered via the Brake Pad Compound. The "Low" setting provides the least friction, reducing the effectiveness of the brakes but allowing the most modulation, while "Medium" and "High" provide more friction and increase the effectiveness of the brakes but allow the least modulation.

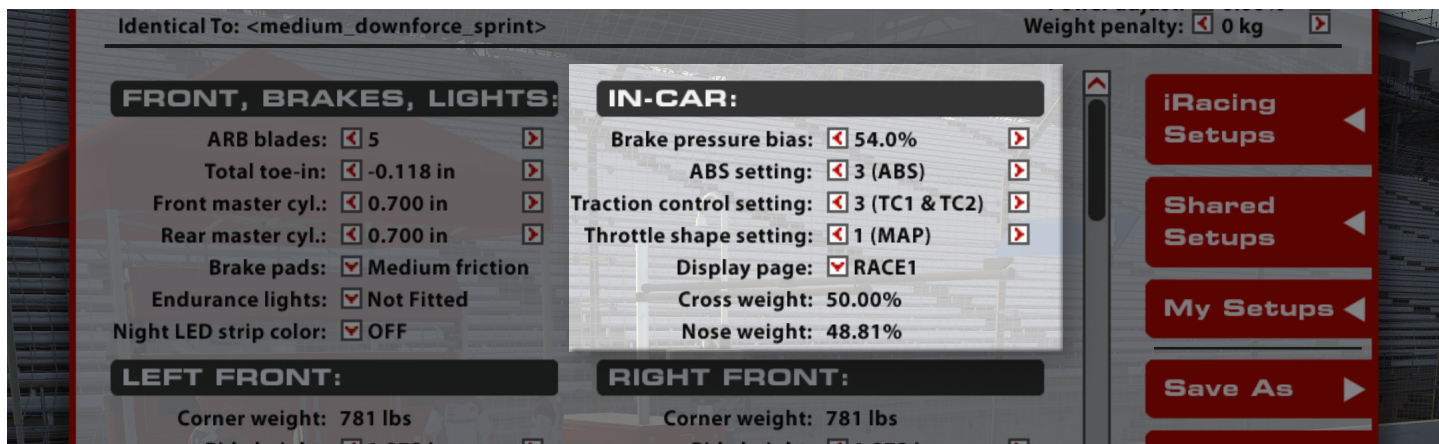
ENDURANCE LIGHTS

An extra set of headlights can be installed for night racing to increase driver visibility. Installing these will not affect vehicle performance.

NIGHT LED STRIP COLOR

The color of LED strips over the doors. These are used to quickly identify the car during a night session.

IN-CAR DIALS



BRAKE PRESSURE BIAS

Brake Bias is the percentage of braking force that is being sent to the front brakes. Values above 50% result in greater pressure in the front brake line relative to the rear brake line which will shift the brake balance forwards increasing the tendency to lock up the front tires but potentially increasing overall stability in braking zones. This should be tuned for both driver preference and track conditions to get the optimum braking performance for a given situation.

ABS SETTING

The current ABS map the car is using. Twelve positions are available: Position 2 has the least intervention/support, position 12 has the most support, and position 1 disables the ABS completely.

More intervention reduces the possibility of and the duration of lockups during braking but can result in longer braking distances if the system is set overly aggressive for the amount of available grip. Positions 2-7 should be used in dry conditions with settings 8-12 for wet conditions. This can be changed in-car via the ABS setting.

TRACTION CONTROL SETTING

The Traction Control setting determines how aggressively the ECU cuts engine torque in reaction to rear wheel spin. Twelve positions are available: Settings 2-12 range from least intervention/sensitivity (position 2) to the highest intervention/sensitivity (position 12) while position 1 disables the traction control completely. More intervention will result in less wheelspin and less rear tire wear but can reduce overall performance if the traction control is cutting engine torque too aggressively and stunting corner exit acceleration. Settings 2-7 are for dry conditions and settings 8-12 are for wet conditions. This can be changed in-car via the TC1 & TC2 settings. TC1 and TC2 are linked together at this time.

THROTTLE SHAPE SETTING

The Throttle Shape setting will adjust how linear the torque delivery is based on the throttle pedal position. There are 3 Settings. Setting 1 is purely linear, with a given percent of throttle delivering a similar percentage of max torque (25% throttle = 25% torque). Setting 2 is a more aggressive, it provides more torque at low throttle angles. Setting 3 is a gradual torque request for use in wet conditions.

DISPLAY PAGE

Changes the currently selected digital dash page. Three options are available as previously described in the dash configuration section of this manual.

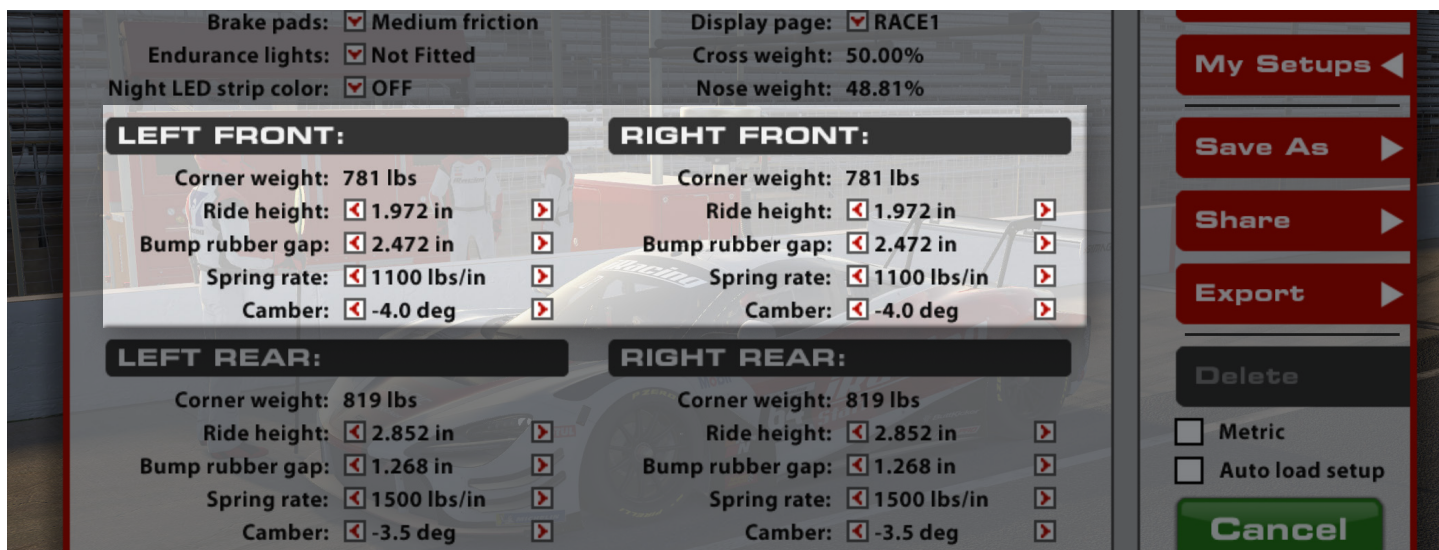
CROSS WEIGHT

The percentage of total vehicle weight in the garage acting across the right front and left rear corners. A setting of 50.0% is generally optimal for non-oval tracks as this will produce symmetrical handling in both left and right hand corners providing all other chassis settings are symmetrical. Higher than 50% cross weight will result in more understeer in left hand corners and increased oversteer in right hand corners. Cross weight can be adjusted by making changes to the ride heights at each corner of the car.

NOSE WEIGHT

The percentage of total vehicle weight on the front two tires. This always includes the driver. It varies with fuel load.

FRONT CORNERS



CORNER WEIGHT

The weight underneath each tire under static conditions in the garage. Correct weight arrangement around the car is crucial for optimizing a car for a given track and conditions. Individual wheel weight adjustments and crossweight adjustments are made via the ride height adjustments at each corner.

RIDE HEIGHT

Distance from the ground to the floor of the car at the front axle centerline. Adjusting Ride Heights is key for optimum performance, as they can directly influence the vehicle's aerodynamic performance as well as mechanical grip. Increasing front ride height will decrease front downforce as well as decrease overall downforce, but will allow for more weight transfer across the front axle when cornering. Conversely, reducing front ride height will increase front and overall downforce, but reduce the weight transfer across the front axle.

BUMP RUBBER GAP

The distance the damper will travel before engaging the bump rubber. This will result in a much stiffer suspension and will provide better aerodynamic platform control and better stability in high-speed corners but it will reduce grip in low-speed corners and over rough surfaces. Lower values will engage the bump rubber sooner and higher values will delay engagement to allow for a more compliant suspension.

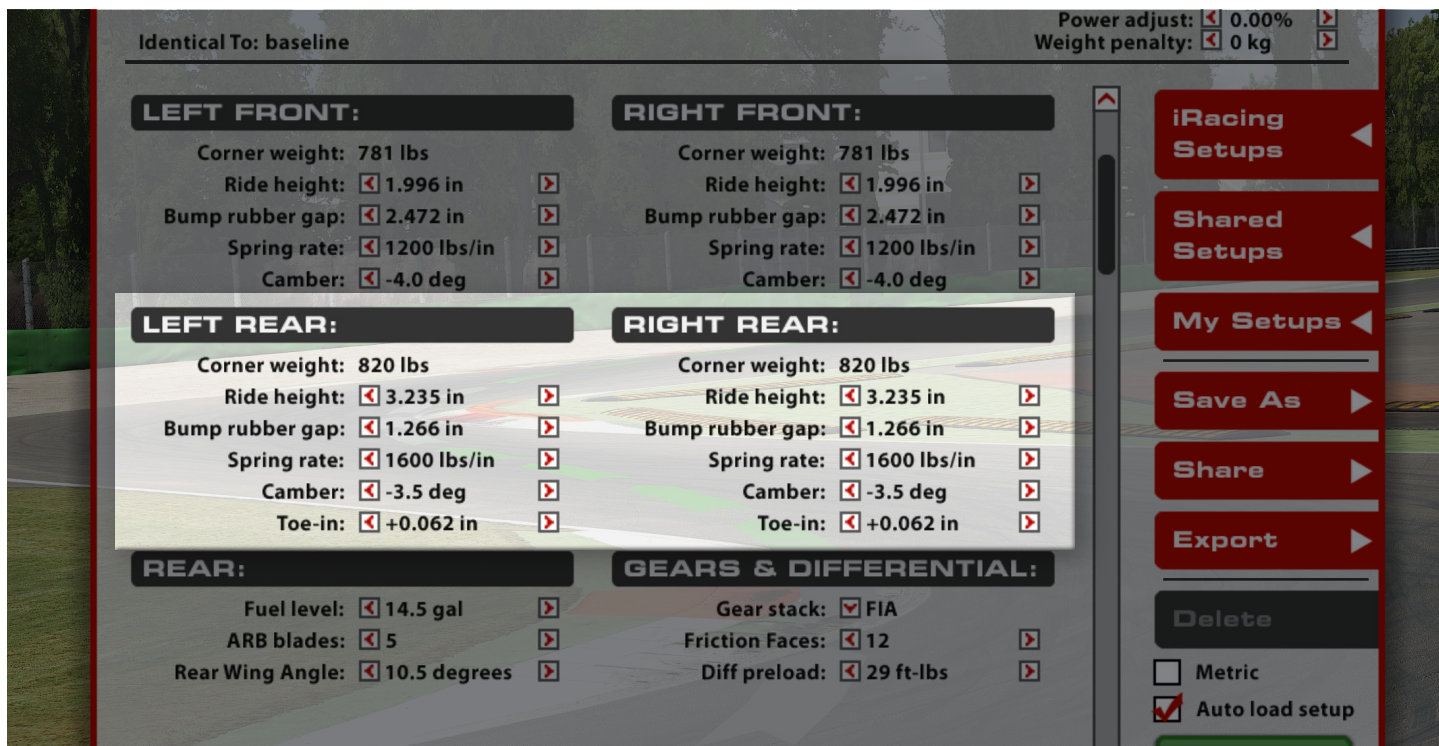
SPRING RATE

This setting determines the installed corner spring stiffness. Stiffer springs will result in a smaller variance in ride height between high and low load cases and will produce superior aerodynamic performance through improved platform control. However overly stiff springs will result in increased tire load variation which will manifest as a loss in mechanical grip. Typically the drawbacks of stiffer springs will become more pronounced on rougher tracks and softer springs in these situations will result in increased overall performance. Corner spring changes will influence both roll and pitch control of the platform and ARB changes should be considered when altering corner spring stiffnesses in order to retain the same front to rear roll stiffness and overall balance. When reducing corner spring stiffness the ARB stiffness should be increased to retain the same roll stiffness as previously.

CAMBER

Camber is the vertical angle of the wheel relative to the center of the chassis. Negative camber is when the top of the wheel is closer to the chassis centerline than the bottom of the wheel, positive camber is when the top of the tire is farther out than the bottom. Due to suspension geometry and corner loads, negative camber is desired on all four wheels. Higher negative camber values will increase the cornering force generated by the tire, but will reduce the amount of longitudinal grip the tire will have under braking. Excessive camber values can produce very high cornering forces but will also significantly reduce tire life, so it is important to find a balance between life and performance. Increasing front camber values will typically result in increased front axle grip during mid to high speed cornering but will result in a loss of braking performance and necessitate a rearward shift in brake bias to compensate.

REAR CORNERS



CORNER WEIGHT

The weight underneath each tire under static conditions in the garage. Correct weight arrangement around the car is crucial for optimizing a car for a given track and conditions. Individual wheel weight adjustments and crossweight adjustments are made by changing individual corner ride heights.

RIDE HEIGHT

Distance from ground to the bottom surface of the car's floor at the rear axle centerline. Increasing rear ride height will decrease rear downforce as well as increase overall downforce and will allow for more weight transfer across the rear axle when cornering. Conversely, reducing ride height will increase rear downforce percentage but reduce overall downforce while reducing the weight transfer across the rear axle. Rear ride height is a critical tuning component for both mechanical and aerodynamic balance considerations and static rear ride heights should be considered and matched to the chosen rear corner springs for optimal performance.

BUMP RUBBER GAP

The distance the damper will travel before engaging the bump rubber. This will result in a much stiffer suspension and will provide better aerodynamic platform control and better stability in high-speed corners but it will reduce grip in low-speed corners and over rough surfaces. Lower values will engage the bump rubber sooner and higher values will delay engagement to allow for a more compliant suspension. Engaging the bump rubbers on the rear can keep the chassis off the track in high-load situations to keep the car from bottoming out on the track, like Daytona, but due to the increased stiffness it can make the car more difficult to control when cornering or during throttle application.

SPRING RATE

Similar to the front axle, stiffer springs will result in a smaller variance in ride height between high and low load cases and will produce superior aerodynamic performance through improved platform control at the expense of mechanical grip. This can be particularly prominent when exiting slow speed corners with aggressive throttle application. Stiffer springs will tend to react poorly during these instances especially so on rough tracks which will result in significant traction loss. Spring stiffness should be matched to the needs of the racetrack and set such that the handling balance is consistent between high and low speed cornering. As an example case, a car which suffers from high speed understeer but low speed oversteer could benefit from an increase in rear spring stiffness. This will allow for a lower static rear height which will reduce rear weight transfer during slow speed cornering while maintaining or even increasing the rear ride height in high speed cornering to shift the aerodynamic balance forwards and reduce understeer.



REAR CORNERS

CAMBER

As with the front of the car it is desirable to run significant amounts of negative camber in order to increase the lateral grip capability; however, it is typical to run slightly reduced rear camber relative to the front. This is primarily for two reasons, firstly, the rear tires are wider compared to the fronts and secondly the rear tires must also perform the duty of driving the car forwards where benefits of camber to lateral grip become a tradeoff against reduced longitudinal (traction) performance.

TOE-IN

Toe is the angle of the wheel, when viewed from above, relative to the centerline of the chassis. Toe-in is when the front of the wheel is closer to the centerline than the rear of the wheel, and Toe-out is the opposite. At the rear of the car it is typical to run toe-in. Increases in toe-in will result in improved straight line stability and a reduction in response during direction changes. Large values of toe-in should be avoided if possible as this will increase rolling drag and reduce straight line speeds. When making rear toe changes remember that the values are for each individual wheel as opposed to paired as at the front. This means that individual values on the rear wheels are twice as powerful as the combined adjustment at the front of the car when the rear toes are summed together. Generally, it is advised to keep the left and right toe values equal to prevent crabbing or asymmetric handling behavior; however, heavily asymmetric tracks such as Lime Rock Park may see a benefit in performance from running asymmetric configurations of rear toe and other setup parameters.

REAR



FUEL LEVEL

The amount of fuel in the fuel tank when the car is loaded into the world.

ARB BLADES

The ARB Blades (or arms) can be adjusted to further tune the suspension roll stiffness beyond only the ARB size setting. This option changes the orientation of the ARB blades and are given numerical values for simplicity, with #1 being the softest option and the blades becoming stiffer as the value is increased to the maximum setting of #7. Stiffer blade settings will increase rear roll stiffness and induce oversteer while softer blade settings will reduce rear roll stiffness and reduce oversteer.

REAR WING ANGLE

The Wing Angle refers to the relative angle of attack of the rear wing, this is an aerodynamic device which has a significant impact upon the total downforce (and drag) produced by the car as well as shifting the aerodynamic balance of the car rearwards with increasing angle. Increasing the rear wing angle results in more total cornering grip capability in medium to high speed corners but will also result in a reduction of straight line speed. Rear wing angle should be adjusted in conjunction with front and rear ride heights, specifically the difference between front and rear ride heights known as 'rake'. To retain the same overall aerodynamic balance it is necessary to increase the rake of the car when increasing the rear wing angle.

GEARS / DIFFERENTIAL



GEAR STACK

Gear Stack changes the forward gear ratios in the transmission. Two choices are available: FIA and IMSA Short. The FIA gear stack is suited to the majority of tracks, including high-speed/low-drag tracks like Daytona and Le Mans. The IMSA Short gear stack is suited to some very high-downforce tracks with top speeds under 240kph.

FRICTION FACES

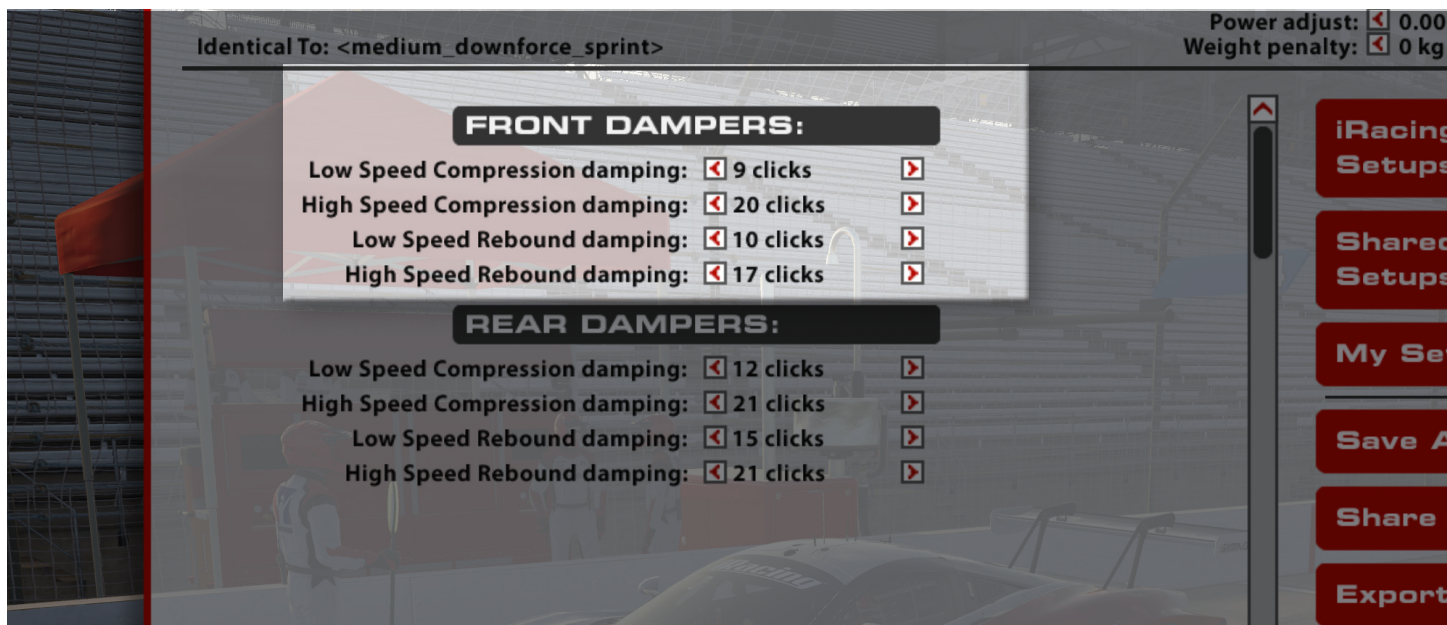
The number of friction faces in the differential affect how much overall force is applied to keep the rear axle locked. Treated as a multiplier, adding more faces produces increasingly more locking force. For example, 8 friction faces will have twice the locking force of 4 faces, which will have twice the force of 2 faces.

DIFFERENTIAL PRELOAD

Diff preload is a static amount of locking force present within the differential and remains constant during both acceleration and deceleration. Increasing diff preload will increase locking on both sides of the differential which will result in more understeer when off throttle and more snap oversteer with aggressive throttle application. Increasing the diff preload will also smooth the transition between on and off throttle behavior as the differential locking force will never reach zero which can be helpful in reducing lift-off oversteer and increasing driver confidence. Typically diff preload should be increased when there is noticeable loss in slow corner exit drive and/or over-rotation during transition between the throttle and brake in low to mid speed corners.

DAMPERS

FRONT DAMPERS



LS COMP DAMPING

Low speed compression affects how resistant the shock is to compression (reduction in length) when the shock is moving at relatively low speeds, usually in chassis movements as a result of driver input (steering, braking, & throttle) and cornering forces. On the McLaren 720S GT3 EVO, setting 40 is minimum damping (least resistance to compression) while setting 0 is maximum damping (most resistance to compression). Increasing the low speed compression damping will result in a faster transfer of weight to this corner of the car during transient movements such as braking and direction change with increased damping usually providing an increase in turn-in response but a reduction in overall grip in the context of front dampers.

HS COMP DAMPING

High speed compression affects the shock's behavior in high speed travel, usually attributed to curb strikes and bumps in the track's surface. Higher compression values will cause the suspension to be stiffer in these situations, while lower values will allow the suspension to absorb these bumps better but may hurt the aerodynamic platform around the track. At smoother tracks more high speed compression damping will typically increase performance while at rougher tracks or ones with aggressive kerbs less high speed compression damping can result in an increase in mechanical grip at the expense of platform control. Setting 0 is maximum damping while Setting 50 is minimum damping.

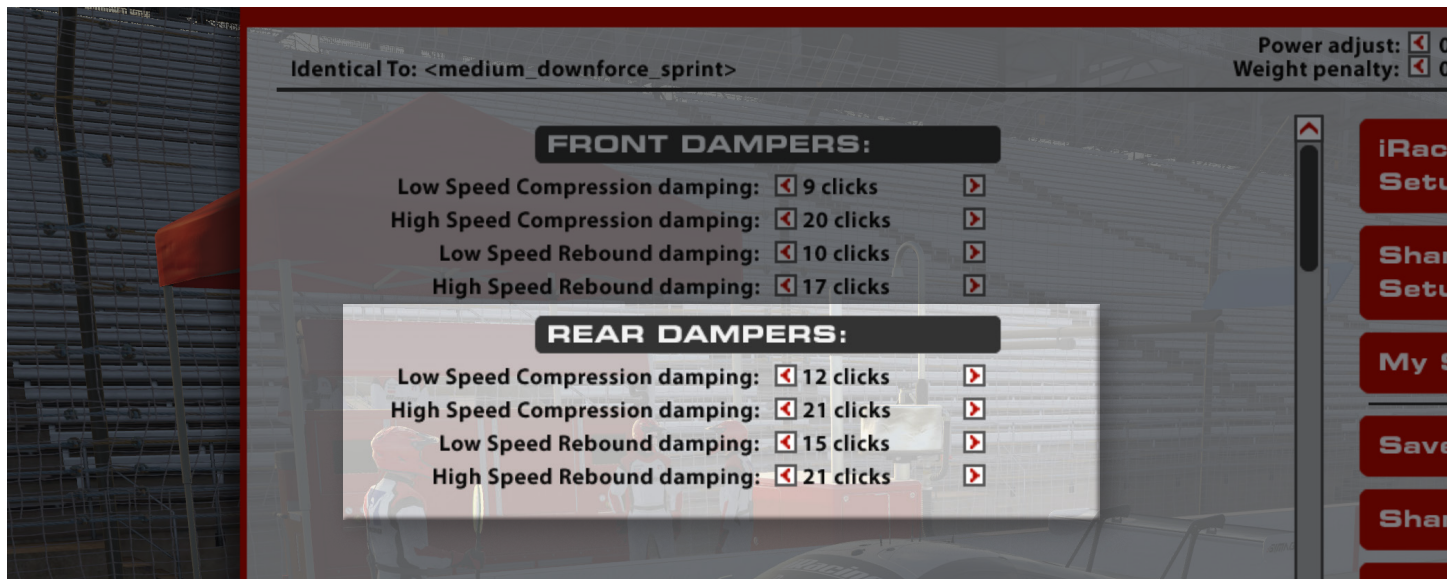
LS RBD DAMPING

Low speed rebound damping controls the stiffness of the shock while extending at lower speeds, typically during body movement as a result of driver inputs. Higher rebound values will resist expansion of the shock, lower values will allow the shock to extend faster. Higher rebound values can better control aerodynamic attitude but can result in the wheel being unloaded when the suspension can't expand enough to maintain proper contact with the track. When tuning for handling, higher front low speed rebound can increase on-throttle mechanical understeer (but reduce nose lift) while lower values will maintain front end grip longer, helping to reduce understeer, but will allow more splitter lift. Excessive front rebound can lead to unwanted oscillations due to the wheel bouncing off of the track surface instead of staying in contact. Setting 0 is maximum damping (most resistant to extension) while Setting 40 is minimum damping (least resistance to extension).

HS RBD DAMPING

High-speed rebound adjusts the shock in extension over bumps and curb strikes. Higher values will reduce how quickly the shock will expand, while lower values will allow the shock to extend more easily. Despite not having as much of an effect on handling in result to driver inputs, High-speed rebound can produce similar results in terms of aerodynamic control and uncontrolled oscillations if set improperly. Setting 0 is maximum damping while Setting 50 is minimum damping.

REAR DAMPERS



LS COMP DAMPING

Low speed compression affects how resistant the shock is to compression (reduction in length) when the shock is moving at relatively low speeds, usually in chassis movements as a result of driver input (steering, braking, & throttle) and cornering forces. On the McLaren 720S GT3 EVO, Setting 40 is minimum damping (least resistance to compression) while Setting 0 is maximum damping (most resistance to compression). Increasing the low speed compression damping will result in a faster transfer of weight to this corner of the car during transient movements such as braking and direction change with increased damping usually increasing the cars tendency to understeer on throttle application.

HS COMP DAMPING

High speed compression affects the shock's behavior in high speed travel, usually attributed to curb strikes and bumps in the track's surface. Higher compression values will cause the suspension to be stiffer in these situations, while lower values will allow the suspension to absorb these bumps better but may hurt the aerodynamic platform around the track. At smoother tracks more high speed compression damping will typically increase performance while at rougher tracks or ones with aggressive kerbs less high speed compression damping can result in an increase in mechanical grip at the expense of platform control. Setting 0 is maximum damping while Setting 50 is minimum damping.

LS RBD DAMPING

Low speed rebound damping controls the stiffness of the shock while extending at lower speeds, typically during body movement as a result of driver inputs. Higher rebound values will resist expansion of the shock, lower values will allow the shock to extend faster. As at the front, high rebound stiffness will result in improved platform control for aerodynamic performance and overall chassis response but it is important to avoid situations where the shock is too slow in rebounding as this can result in the tire losing complete contact with the track surface. Provided this is avoided,, an increase in rebound stiffness can help to 'slow down' the change in pitch of the car as the brakes are applied, increasing braking stability and off-throttle mechanical understeer. Setting 0 is maximum damping (most resistant to extension) while Setting 40 is minimum damping (least resistance to extension).

HS RBD DAMPING

High-speed rebound adjusts the shock in extension over bumps and curb strikes. Higher values will reduce how quickly the shock will expand, while lower values will allow the shock to extend more easily. Despite not having as much of an effect on handling in result to driver inputs, High-speed rebound can produce similar results in terms of aerodynamic control and uncontrolled oscillations if set improperly. Setting 0 is maximum damping while Setting 50 is minimum damping.

SETUP TIPS

This section is aimed toward helping users who want to dive deeper into the different aspects of the vehicle's setup.



SETUP TIPS

Baseline is a 50% fuel load setup which is intended solely for loading the car. As such, this setup should always pass tech inspection at every fuel load and track (Except Nürburgring Nordschleife configurations where 'nuburgring_sprint/endurance' should be used) but will not provide ultimate performance.

Setups labeled '_wet' have wet tyres pre-fitted and setup adjustments to suit wet conditions.

Setups labeled '_sprint' have a 50% fuel load, a more aggressive balance and are intended for use where there is either a fuel limitation OR race lengths are approximately 25 to 30 minutes in length. These setups are intended to be used in competition.

Setups labeled '_endurance' have a 100% fuel load and are for use where no fuel restriction is present and/or race lengths are approximately 1 hour or more in length.

The setup titled 'fixed' is the setup used in the fixed setup series and is similar to the high_downforce_sprint setup.

Setups labeled 'nurburgring_' are built with 70 mm minimum ride heights and are for use solely on Nürburgring Nordschleife configurations.

While most tracks will trend towards favoring more downforce there can be some instances where reducing rear wing angle for less drag may be beneficial. As a rough guide, you can expect the following downforce trims at the following tracks:

Tracks	Downforce Level	Tracks	Downforce Level
Autodromo Jose Carlos Pace	High/Medium	Long Beach Street Circuit	High
Autodromo Nazionale Monza	Medium	Motorsports Arena Oschersleben	High
Brands Hatch Circuit	High	Mount Panorama Circuit	High/Medium
Circuit de Barcelona Catalunya	High	Nürburgring Grand-Prix-Strecke	High
Circuit de Nevers Magny-Cours	High/Medium	Okayama International Circuit	High
Circuit de Spa-Francorchamps	Medium	Road America	High/Medium
Circuit des 24 Heures Du Mans	Medium	Sebring International Raceway	High
Daytona International Speedway	Low/Medium	Silverstone Circuit	High/Medium
Detroit Grand Prix at Belle Isle	High	Sonoma Raceway	High
Fuji International Speedway	High/Medium	Virginia International Raceway	High/Medium
Hungaroring	High	Watkins Glen International	High/Medium
Indianapolis Motor Speedway	Medium	WeatherTech Raceway at Laguna Seca	High
Lime Rock Park	High		

Should you wish to drive at a track not listed it is recommended to start out with the High Downforce setup first before evaluating the other downforce level options. A good indicator of if a track may benefit from a reduction in downforce trim is the maximum speed reached.

The following boundaries are suggestions for what trim level may be optimal but please note that other factors such as track design (number of high speed corners, etc), altitude and ambient conditions will also impact your decision here with higher altitude tracks and hotter ambient conditions favoring more downforce.

Speed	Downforce Level
Max Speed under 250 km/h (155 mph)	High Downforce
Max Speed 250 to 270 km/h	Medium
Max Speed over 270 km/h (167 mph)	Low to Minimum Downforce

AERODYNAMIC TARGETS AND ADJUSTMENTS

GT3 cars are very sensitive to small variations in ride heights at both the front and rear axle and this must be kept in mind when making setup adjustments such as static ride heights, corner spring rates and rear wing angle.

The optimal configuration for most total downforce is as follows:

Rear Wing Angle: +10.5

Dynamic Front Ride Height: 35.0 mm (+/-2.5 mm)

Dynamic Rear Ride Height: 70.0 mm (+/-2.5 mm)

Should you go over or under the ride height targets stated above you will begin to lose overall downforce. It is very important to consider all aspects of the track when aiming for this maximum downforce target. Consider that if the rear ride height increases beyond the target during braking, you will experience both a balance shift forwards and a loss in overall downforce resulting in a destabilizing situation. It is these braking considerations that will govern how closely you can approach this maximum in a real world situation.

The optimal configuration for the least total drag is as follows:

Rear Wing Angle: +0.5

Dynamic Front Ride Height: 17.5 mm (+/- 2.5 mm)

Dynamic Rear Ride Height: 17.5 mm (+/- 2.5 mm)

For the majority of tracks, it will be difficult to achieve ride heights low enough to hit these drag targets; however, it is possible at a track such as Daytona. Please keep in mind that your absolute minimums are governed by the road surface and that while aerodynamic drag will decrease as you approach these targets, overall drag may increase if the car starts to make ground contact. It should also be stated that this low drag trim is neither optimal for total downforce nor handling balance.

When adjusting the rear wing angle, the following adjustments should be made to retain aerodynamic balance:

Rear Wing Angle: +1

Front Ride Height: -1.5 mm

OR

Rear Ride Height: +4.5 mm

Rear Wing Angle: -1

Front Ride Height: +1.5 mm

OR

Rear Ride Height: -4.5 mm

These adjustment sensitivities are not the same in all parts of the aeromap nor are they valid for all rear wing angles. For this reason, it is strongly suggested that one use the Aero Balance Calc tool on the Tires/Aero tab:

- Start with a setup that is known to have good high speed aero balance.
- Note the % Front Downforce and dynamic RHs at speed.
- Add or subtract Rear Wing Angle for the desired overall downforce change
- Adjust the Rear RH at speed until the target % Front Downforce value is reached
- Apply the difference in dynamic Rear RH needed to retain balance to the static Rear RHs on the Chassis tab.

It is also possible to combine adjustments of front and rear ride height together if necessary (such as when lower rear heights cannot be easily achieved), this can result in more overall downforce being retained when reducing wing angle without detrimentally impacting the balance but at the cost of slightly increased aerodynamic drag.

These reference values are provided as targets to aim for, however, overall car balance should remain the priority. It may not be possible to achieve a good balance at these targets in certain situations and as such, you should elect to sacrifice some raw performance for a better balance.

Lower Rear Wing Angle = More oversteer, less downforce, less drag, lower cornering speed, higher straight line speed.

Higher Rear Wing Angle = More understeer, more downforce, more drag, higher cornering speed, lower straight line speed.



CHASSIS ADJUSTMENTS

Should you wish to adjust the underpinning balance of the car without impacting the aero platform significantly in pitch and heave, or adjusting the differential then front and rear adjustable anti-roll bars are available.

Stiffer front ARB → More Understeer

Softer front ARB → More Oversteer

Stiffer rear ARB → More Oversteer

Softer rear ARB → More Understeer

Softer front AND rear ARB → Reduced aerodynamic performance, more mechanical grip (good for rough surfaces) and slower response to inputs.

Stiffer front AND rear ARB → Increased aerodynamic performance (good for fast sweeping corners), less mechanical grip and increased response to inputs.

DIFFERENTIAL ADJUSTMENTS

Two adjustment options are available for the differential.

More friction faces → More off throttle understeer, more on throttle oversteer, less inside wheelspin-up on rough surfaces and kerb strikes.

Less friction faces → Less off throttle understeer, less on throttle oversteer, more inside wheelspin-up on rough surfaces and kerb strikes. Typically better at tracks like Spa or those with smooth surfaces and flat kerbing.

Friction faces are dominant at high input torques such as full throttle, sustained braking or pure coastdown.

Preload is additive to the total locking torque of the differential and acts as an offset torque which is always present, even at zero input torque. This means that it is more dominant during transition behavior where the differential input torque is near zero, such as at throttle lift and/or during initial trail braking.

More preload → Less liftoff oversteer, more corner entry stability, more off throttle understeer, more on throttle oversteer.

Less preload → More liftoff oversteer, less corner entry stability, less off throttle understeer, less on throttle oversteer.